Project 2 — Stochastic Modelling

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November 4, 2026

1 Problem 2: Calibrating Climate Models

We model the unknown relationship between the parameter θ and the score using a Gaussian process.

1.1 a) Prediction

The expectation and variance are given by

$$E[Y(\theta)] = 0.5, \qquad \text{Var}[Y(\theta)] = 0.5^2.$$

Here we describe how we calculated the conditional distribution.

1.2 b) Probability of Reaching the Target

Here we describe our method and results.

1.3 c) A New Observation

Here we discuss how the new observation affects the prediction.